# Ongoing Work on Contextual Neural Machine Translation


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# Motivation

- **Most neural machine translation (NMT) systems translate sentence by sentence**

- **Make use of surrounding sentences to exploit context information**
  - **reduce ambiguities in translation choices**
  - **example: cross-sentence anaphora in English→German**

| Source sentences | Correct translation of 'it' |
|---|---|
| *The monkey ate the banana. Because it was hungry.* | er |
| *The monkey ate the banana. Because it ripe.* | sie |
| *The monkey ate the banana. Because it was tea-time.* | es |

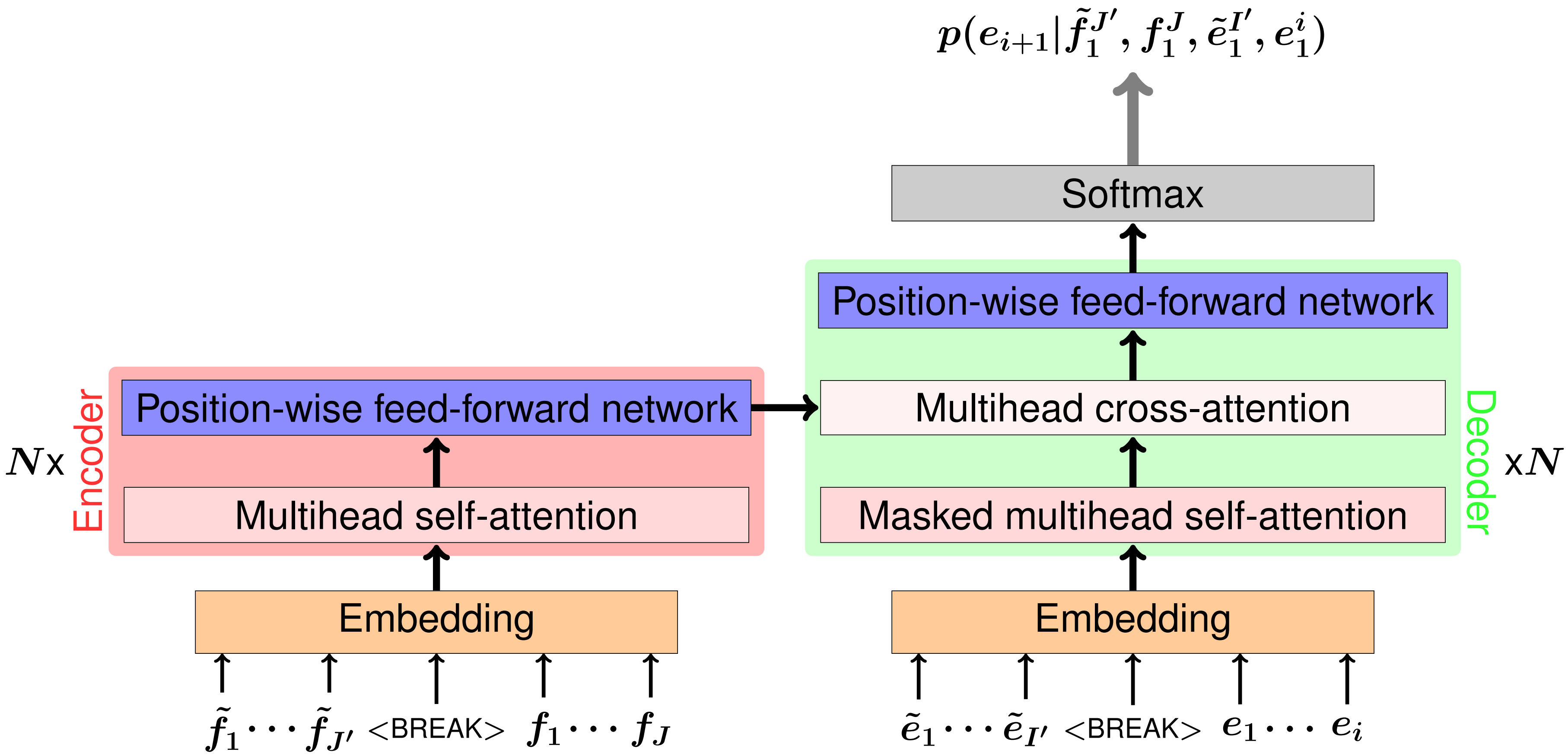
# State of the Art Literature

- Single encoder: concatenate sentences on source-/target side [Tiedemann & Scherrer 17, Agrawal & Turchi$^{+}$ 18]

- Multi-Encoder with attention mechanism
  - Feed surrounding sentences into seperate encoders and combine context vectors:
    - concatenation (like in [Zoph & Knight 16])
    - gated sum [Bawden & Sennrich$^{+}$ 18]
    - hierarchical attention [Bawden & Sennrich$^{+}$ 18]

**Remark: above methods only consider *local* context**

# Transformer Architecture with Concatenated Context Sentences




- **one previous context-sentence on the source ($\tilde{f}_1^{J'}$) and target ($\tilde{e}_1^{I'}$) side**
- **<BREAK> token seperates the sentences**

# Ongoing Work at i6

1. Reproduce results from [Agrawal & Turchi$^{+}$ 18]
   - Transformer model
   - IWSLT 2017 English$\rightarrow$Italian
   - concatenate sentences with special token <BREAK>, example: `sentence_1` <BREAK> `sentence_2`
2. Compress context-sentences by pruning
   - first $n$ most-common words in training corpus
   - pre-defined stopword list
3. Test method of concatenation (+ pruning) on WMT 2018 German$\rightarrow$English data as well

# Concatenation

(a) Single target sentence.

| Source | $\rightarrow$ | Target |
|---|---|---|
| 1 previous + current | $\rightarrow$ | current |
| 2 previous + current | $\rightarrow$ | current |
| 1 previous + current + 1 next | $\rightarrow$ | current |
| current | $\rightarrow$ | current |

(b) Multiple target sentence.

| Source | $\rightarrow$ | Target |
|---|---|---|
| 1 previous + current | $\rightarrow$ | 1 previous + current |
| 1 previous + current + 1 next | $\rightarrow$ | 1 previous + current |

# Data Statistics

| | | IWSLT'17 EN-IT | | WMT'18 DE-EN | |
|---|---|---|---|---|---|
| | | EN | IT | DE | EN |
| Train. | Sentences | 232K | | 5.9M | |
| | Running words | 4.7M | 4.4M | 160.9M | 157.7M |
| Dev. | | *tst2010* | | *newstest2016* | |
| | Sentences | 1.5K | | 3K | |
| | Running words | 31K | 29K | 77K | 73K |
| Test | | *tst2017* | | *newstest2017* | |
| | Sentences | 1.1K | | 3K | |
| | Running words | 22K | 20K | 76K | 73K |

# Results: IWSLT 2017 English→Italian

Evaluation results on the test set with concatenation ( + ). We report de-tokenized and cased BLEU scores.

| | | | *tst2017* | |
|---|---|---|---|---|
| | | | BLEU [%] | TER [%] |
| current | → | current | 29.89 | 52.0 |
| 1 previous + current | → | current | 30.02 | 51.8 |
| 2 previous + current | → | current | 29.81 | 52.2 |
| 1 previous + current + 1 next | → | current | 29.32 | 52.8 |
| current + 1 next | → | current | 29.93 | 51.9 |
| 1 previous + current | → | 1 previous + current | 30.64 | 51.0 |
| 1 previous + current + 1 next | → | 1 previous + current | 30.64 | 51.4 |

*Single* target sentence.

| Source | → Target | tst2017 BLEU [%] | tst2017 TER [%] |
|---|---|---|---|
| current | → current | 29.89 | 52.0 |
| 1 previous + current | → current | 30.02 | 51.8 |
| + with most-common-100 pruning (59% pruned words) | | 30.03 | 52.0 |
| 2 previous + current | → current | 29.81 | 52.2 |
| + with most-common-90 pruning (57% pruned words) | | 30.35 | 51.4 |
| 1 previous + current + 1 next | → current | 29.32 | 52.9 |
| + with most-common-90 pruning (57% pruned words) | | 30.14 | 51.8 |
| current + 1 next | → current | 29.93 | 51.9 |
| + with most-common-100 pruning (59% pruned words) | | 30.00 | 51.6 |

# Results: IWSLT 2017 English→Italian

*Multiple* target sentences.

| Source → Target | tst2017 BLEU [%] | tst2017 TER [%] |
|---|---|---|
| 1 previous + current → 1 previous + current | 30.64 | 51.0 |
| + with most-common-90 pruning (57% pruned words) | 30.09 | 52.0 |
| 1 previous + current + 1 next → 1 previous + current | 30.64 | 51.4 |
| + with most-common-150 pruning (64% pruned words) | 30.18 | 51.6 |

# Results: WMT 2018 German→English

| | | | *newstest2017* | |
|---|---|---|---|---|
| | | | BLEU [%] | TER [%] |
| current | → | current | 30.36 | 51.0 |
| 1 previous + current | → | current | 30.13 | 51.4 |
| 2 previous + current | → | current | 30.32 | 51.2 |
| 1 previous + current + 1 next | → | current | 30.28 | 51.2 |
| current + 1 next | → | current | 30.15 | 51.0 |
| 1 previous + current | → | 1 previous + current | 30.10 | 51.2 |
| 1 previous + current + 1 next | → | 1 previous + current | 30.18 | 51.2 |

- [another slide with most-common pruning - still in training]

# Summary and Future Work

- Concatenation of sentences yield slight performance increases with regards to BLEU and TER metrics
  - preliminary results on WMT 2018 German→English
- Simple compression by pruning words according to frequencies
  - IWSLT data set: up to +1 BLEU improvement
  - WMT data set: ► [check values]

**Future work:**

- Implementation of multi-encoder approaches [Bawden & Sennrich+ 18, Voita & Sennrich+ 18]
- Test approaches in discourse-heavy data sets like OpenSubtitles [Lison & Tiedemann 16]
- Consider global context information: e.g. topic embedding

# Thank you for your attention

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